

Annie Prasanna Manoharan

+1 (479) 301-1366 | annieprm0501@gmail.com | annie0159.github.io/Portfolio | github.com/Annie0159

Education

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College, India

Aug 2019 – May 2023

Bachelor of Engineering in Electronics and Communication

Technical Skills

Programming: Python, SQL, Java, C++, C, C#

Data Engineering: ETL, ELT, Data Pipelines, Data Ingestion, Transformation, Validation, Data Quality, Batch Processing, Real-Time Processing, Workflow Automation

Streaming & Distributed: Apache Kafka, Apache Flink, Flink SQL, Event-Time Processing, Windowing, Watermarking, Apache Spark, PySpark, Delta Lake, Databricks

Orchestration & Cloud: Apache Airflow, DAGs, AWS Glue, Lambda, EMR, S3, EC2, RDS, Azure SQL, Docker, Linux

Databases: SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, ClickHouse, OLAP, Data Modeling, Schema Design

Tools: Git, GitHub, Bitbucket, Jira, Grafana, Streamlit, Agile, Scrum, SDLC

Professional Experience

Data Engineering Intern, InsOps Inc.

Jul 2024 – Present

- Developed data pipelines in AWS using Glue, Lambda, EMR, SQL and Apache Airflow for orchestration, with S3 storage, for scalable data ingestion, transformation, validation and analytics
- Built data workflows integrating SQL Server, Azure SQL and AWS S3, preparing structured datasets for downstream data science and analytics
- Implemented metadata-driven data processing for automated schema discovery, transformation and data validation across heterogeneous sources
- Worked with distributed processing and lakehouse concepts, including scalable ETL/ELT, cloud data storage, SQL transformations and pipeline orchestration

Teaching Assistant, Dept. of Computer Science & Engineering, University of Arkansas

Sep 2024 – May 2026

- Supported Database Management Systems instruction for approximately 180 students, assisting with database concepts, assignments, assessments and technical questions
- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance and mentorship in C++ fundamentals

Project Engineer, Soliton Technologies, India

Jun 2023 – Jun 2024

- Designed and developed a C# WPF desktop application integrating C#, Python and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Medtronic as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Collaborated with cross-functional engineering teams to debug system-level issues, optimize workflows and deliver production-ready software

Projects

Real-Time Medical IoT Analytics Platform

Apache Kafka, Apache Flink, SQL, Python, ClickHouse, Grafana

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process and score high-frequency medical IoT device data
- Implemented temporal Flink OVER windows partitioned by device ID to compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to handle out-of-order streams
- Designed ClickHouse schemas using the MergeTree engine sorted by composite keys to align disk layout with clinical query patterns, isolating ingestion via an optimized Python multi-topic batch consumer

Local LLM-Powered News Research Platform

Python, LangChain, Streamlit, FAISS, Ollama

- Built an end-to-end local ETL pipeline using Streamlit and WebBaseLoader to ingest, clean, chunk and preprocess unstructured web content on demand
- Designed a local vector storage architecture using FAISS and nomic-embed-text, and a retrieval-augmented generation pipeline via LangChain over local Llama 3.2 instances